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Volume 01 — Essentials, APT/Kill Chain, Firewalls (L01–L09)

> Cyber Security Tools · IGNOU/BAOU nou26_ge86 · Prof. Dr. Nilesh K. Modi
> **notes-3** — concept exam notes (what things are / what they do). Condensed from course `pdf/` volumes. No lab command dumps.

Tool landscape, CIA/risk, Nmap/Netcat/Wireshark roles, Kill Chain, STRIDE, firewall concepts.

L1: Course Introduction & Cyber Security Tool Landscape

What it is / what it does

- Security tools are software utilities used to **detect**, **prevent**, **analyze**, or **respond to** cyber threats. This course maps tools to the BAOU PGDCS-103 blocks and hands-on lab workflows.
- **Tool categories:**

Category	Examples	Role
Reconnaissance	Nmap, theHarvester	Discover assets
Scanning	Nikto, OpenVAS	Find vulnerabilities
Exploitation	sqlmap, Hydra	Demonstrate impact (lab only)
Defense	iptables, Snort, Windows Firewall	Block/detect attacks
Analysis	Wireshark, strings	Inspect traffic/files

Defenses (one-liners)

- Maintain asset inventory so unknown scanning is detectable.
- Segment lab networks from production LANs.
- Enforce **authorization** before any security testing.
- Log and monitor VM host activity.

Exam lines

- Security tools are software utilities used to **detect**, **prevent**, **analyze**, or **respond to** cyber threats. This course maps…
- **Tool categories:**

L2: CIA Triad, Risk Assessment & Security Policies

What it is / what it does

- Establish the security foundation before tool usage. Aligns with PGDCS-101 Block 1 and Nina Godbole's risk-management chapters.
- **Confidentiality** — encryption, access control
- **Integrity** — hashing, change detection
- **Availability** — redundancy, backups, DDoS mitigation
- **Policy** — what must be done (e.g., 'All passwords ≥ 12 characters').
- **Procedure** — how to do it (e.g., Active Directory GPO steps).

Defenses (one-liners)

- Publish acceptable-use and password policies.
- Conduct annual risk assessments.
- Map controls to frameworks (ISO 27001, NIST).

Exam lines

- Establish the security foundation before tool usage. Aligns with PGDCS-101 Block 1 and Nina Godbole's risk-management chapters.
- **Confidentiality** — encryption, access control
- **Integrity** — hashing, change detection
- **Availability** — redundancy, backups, DDoS mitigation
- **Policy** — what must be done (e.g., 'All passwords ≥ 12 characters').

L3: Network Discovery with Nmap & Netcat

What it is / what it does

- **Nmap:** Network mapper for host discovery, port scanning, service/version detection, and OS fingerprinting.
- **Netcat:** Swiss-army TCP/UDP tool for connectivity testing, banner grabbing, and simple file transfer.

Defenses (one-liners)

- Deploy IDS/IPS to detect SYN scans (nmap -sS).
- Close unused ports; firewall by default-deny.
- Use port knocking or jump hosts for admin access.

- Rate-limit connection attempts (fail2ban).

Exam lines

- **Nmap:** Network mapper for host discovery, port scanning, service/version detection, and OS fingerprinting.
- **Netcat:** Swiss-army TCP/UDP tool for connectivity testing, banner grabbing, and simple file transfer.

L4: Packet Analysis with Wireshark

What it is / what it does

- Capture and decode network frames to troubleshoot issues and detect malicious traffic.
- **Key display filters:**

Filter	Use
ip.addr == 10.0.0.5	Host traffic
tcp.flags.syn == 1	SYN packets
dns	DNS queries
tls.handshake.type == 1	TLS ClientHello

Defenses (one-liners)

- Encrypt traffic (TLS 1.2+) to limit passive disclosure.
- Monitor for DNS tunneling and unusual protocols.
- Store PCAPs securely with access controls.
- Use encrypted management channels (SSH, not Telnet).

Exam lines

- Capture and decode network frames to troubleshoot issues and detect malicious traffic.
- **Key display filters:**

L5: APT Lifecycle & Cyber Kill Chain

What it is / what it does

- Understand how advanced persistent threats progress through stages — maps offensive tools to defensive detection points.
- **Reconnaissance** — OSINT, Nmap
- **Weaponization** — malware + exploit paired
- **Delivery** — phishing email, drive-by

- **Exploitation** — trigger vulnerability
- **Installation** — persistence, backdoor
- **Command & Control** — beacon to C2 server
- **Actions on Objectives** — data theft, destruction
- **MITRE ATT&CK mapping** — Use ATT&CK Navigator to map techniques (T1595 Recon, T1190 Exploit Public-Facing App) to tools covered in this course.

Defenses (one-liners)

- **Defense in depth** — block multiple stages.
- Email filtering + user awareness (break delivery).
- EDR for installation/C2 detection.
- Network segmentation to limit lateral movement.

Exam lines

- Understand how advanced persistent threats progress through stages
- **Reconnaissance** — OSINT, Nmap
- **Weaponization** — malware + exploit paired
- **Delivery** — phishing email, drive-by
- **MITRE ATT&CK mapping** — Use ATT&CK Navigator to map techniques (T1595 Recon,…

L6: Threat Modeling & Attack Surface Mapping

What it is / what it does

- Systematically identify threats before attackers do; prioritize which tools to deploy for testing.
- **STRIDE model:**

Threat	Example	Tool to test
Spoofing	Fake login page	Phishing sim
Tampering	SQL injection	sqlmap
Repudiation	Log deletion	auditd
Info disclosure	Verbose errors	Nikto
DoS	SYN flood	hping3 (lab)
Elevation	SUID exploit	manual

Defenses (one-liners)

- Reduce attack surface: disable unused services.
- Apply least privilege.
- Re-model after every architecture change.
- Integrate threat modeling into SDLC.

Exam lines

- Systematically identify threats before attackers do; prioritize which tools to deploy for testing.
- **STRIDE model:**

L7: Firewall Concepts — Stateful vs Stateless Filtering

What it is / what it does

- Firewalls enforce access control at network boundaries — foundation for iptables (L8) and Windows Firewall (L9).
- **Packet-filtering** — examines headers (IP, port); stateless.
- **Stateful** — tracks connection state; allows return traffic automatically.
- **Application-layer (proxy)** — inspects HTTP/DNS content.
- **Next-gen (NGFW)** — IPS, app awareness, threat intel.
- **Default deny** — block all, allow explicitly.
- **Least privilege** — minimum ports/services.
- **Document** every rule with owner and expiry.
- Test rules in staging before production.

Defenses (one-liners)

- Place firewalls at trust boundaries (DMZ, internal segments).
- Log denied packets; alert on spikes.
- Combine with IDS (Snort) for deep inspection.
- Review rules quarterly; remove orphans.

Exam lines

- Firewalls enforce access control at network boundaries
- **Packet-filtering** — examines headers (IP, port); stateless.
- **Stateful** — tracks connection state; allows return traffic automatically.
- **Application-layer (proxy)** — inspects HTTP/DNS content.
- **Default deny** — block all, allow explicitly.

L8: Linux Packet Filtering with iptables & UFW

What it is / what it does

- **iptables:** Linux kernel netfilter framework for packet filtering, NAT, and mangling.

Defenses (one-liners)

- Automate rule deployment (Ansible, Puppet).
- Never expose management ports to the internet.
- Test with nmap after rule changes.

- Monitor /var/log/kern.log for DROP entries.

Exam lines

- **iptables:** Linux kernel netfilter framework for packet filtering, NAT, and mangling.

L9: Windows Firewall & Host-Based Perimeter Defense

What it is / what it does

- Windows Defender Firewall provides host-level inbound/outbound filtering on Windows workstations and servers.

Defenses (one-liners)

- Enforce GPO-managed firewall rules in AD environments.
- Combine with Defender ATP/EDR.
- Disable unnecessary services (SMBv1, Telnet).
- Audit rule changes via Windows Event Log (4946, 4947).

Exam lines

- Windows Defender Firewall provides host-level inbound/outbound filtering on Windows workstations and servers.